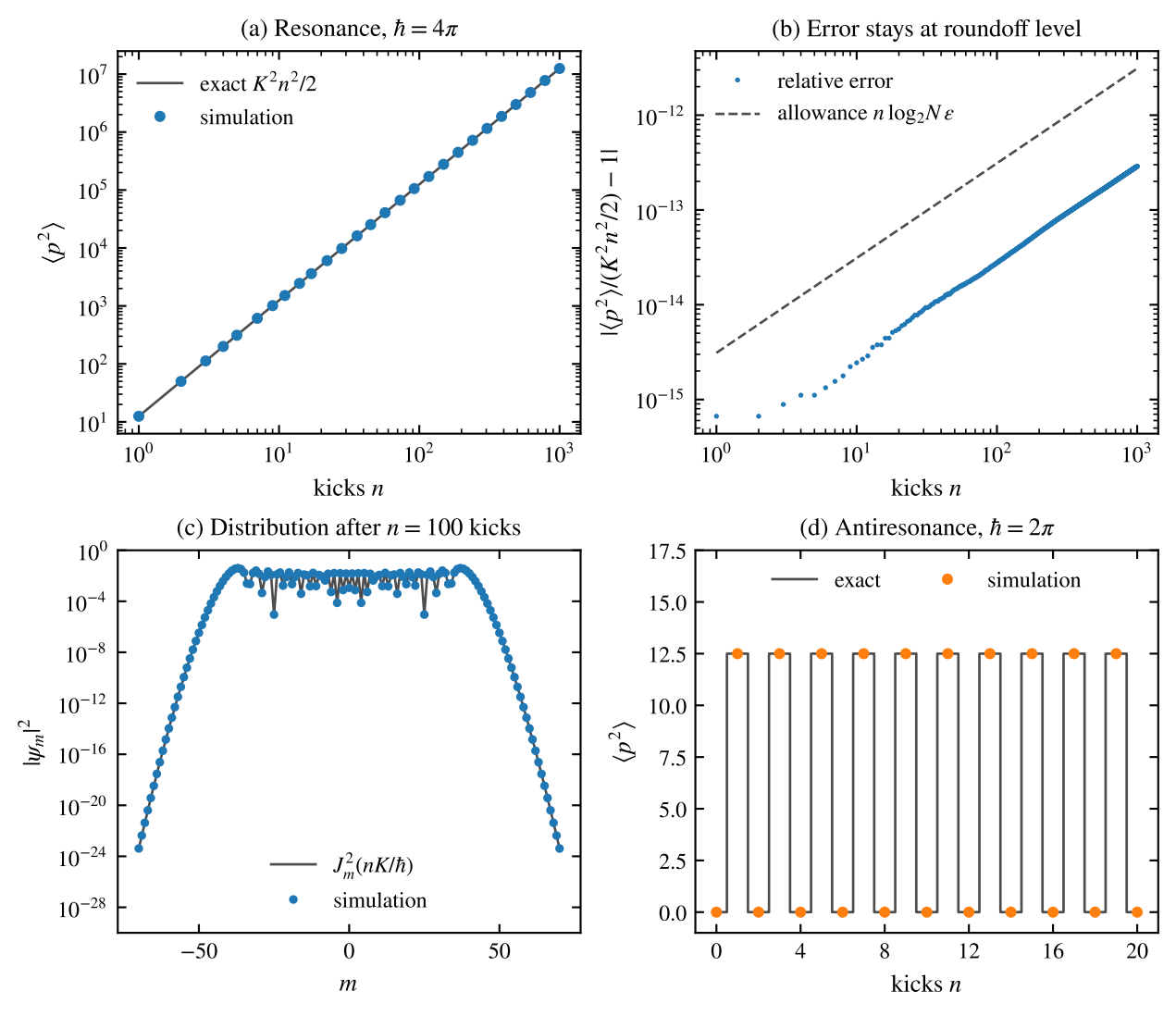
(a) Resonance, $\hbar = 4\pi$
exact $K^2n^2/2$
simulation
$\langle p^2 \rangle$
kicks n
(b) Error stays at roundoff level
relative error
allowance $n \log_2 N \epsilon$
$|\langle p^2 \rangle/(K^2n^2/2) - 1|$
kicks n
(c) Distribution after $n = 100$ kicks
$J_m^2(nK/\hbar)$
simulation
$|\psi_m|^2$
m
(d) Antiresonance, $\hbar = 2\pi$
exact
simulation
$\langle p^2 \rangle$
kicks n